Benzene and Aromatic Compounds

Structure, stability and substitution

Name _____ Class _____ Date _____

DATA

Bond lengths: C–C 154 pm · C=C 134 pm · ΔH(hydrogenation): cyclohexene −120 kJ/mol, benzene −208 kJ/mol

Molar masses (g/mol): C 12.01 · H 1.008 · C_6H_6 78.11 · C_6H_5Br 157.01 · C_7H_8 92.13 · $C_7H_7NO_2$ 137.14

Percent yield = actual ÷ theoretical × 100 %

Show units in every calculation step.

A. Structure

1. (a) Draw the Kekulé structure and the delocalized structure of benzene. (b) Describe the bonding in benzene in terms of hybridization and p orbitals.

2. Give two pieces of evidence that benzene does not contain three ordinary C=C bonds.

3. Benzene is 92.3 % carbon by mass. Show that its empirical formula is CH.

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B. Stability and reactions

4. Use the hydrogenation data to calculate the delocalization energy of benzene, and explain what it means.

5. Give the reagents, conditions and product for (a) the nitration of benzene (b) the bromination of benzene. Name the type of reaction.

6. Name: (a) $C_6H_5CH_3$ (b) C_6H_5COOH (c) C_6H_5Cl (d) C_6H_5OH

C. Calculations

7. 15.6 g of benzene is converted completely into bromobenzene. Calculate the theoretical mass of bromobenzene.

D. Challenge

8. 9.21 g of methylbenzene (C_7H_8) is nitrated to give a single product, $C_7H_7NO_2$. 10.5 g of product is obtained. Calculate the theoretical yield and the percent yield.

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